SCIENCE MASTER PROGRAMME

SCIENCE MASTER MANUAL

Chemistry • Biology • Physics

A concise learning and question–answer manual covering important basic concepts of Science.

PART 1
CHEMISTRY

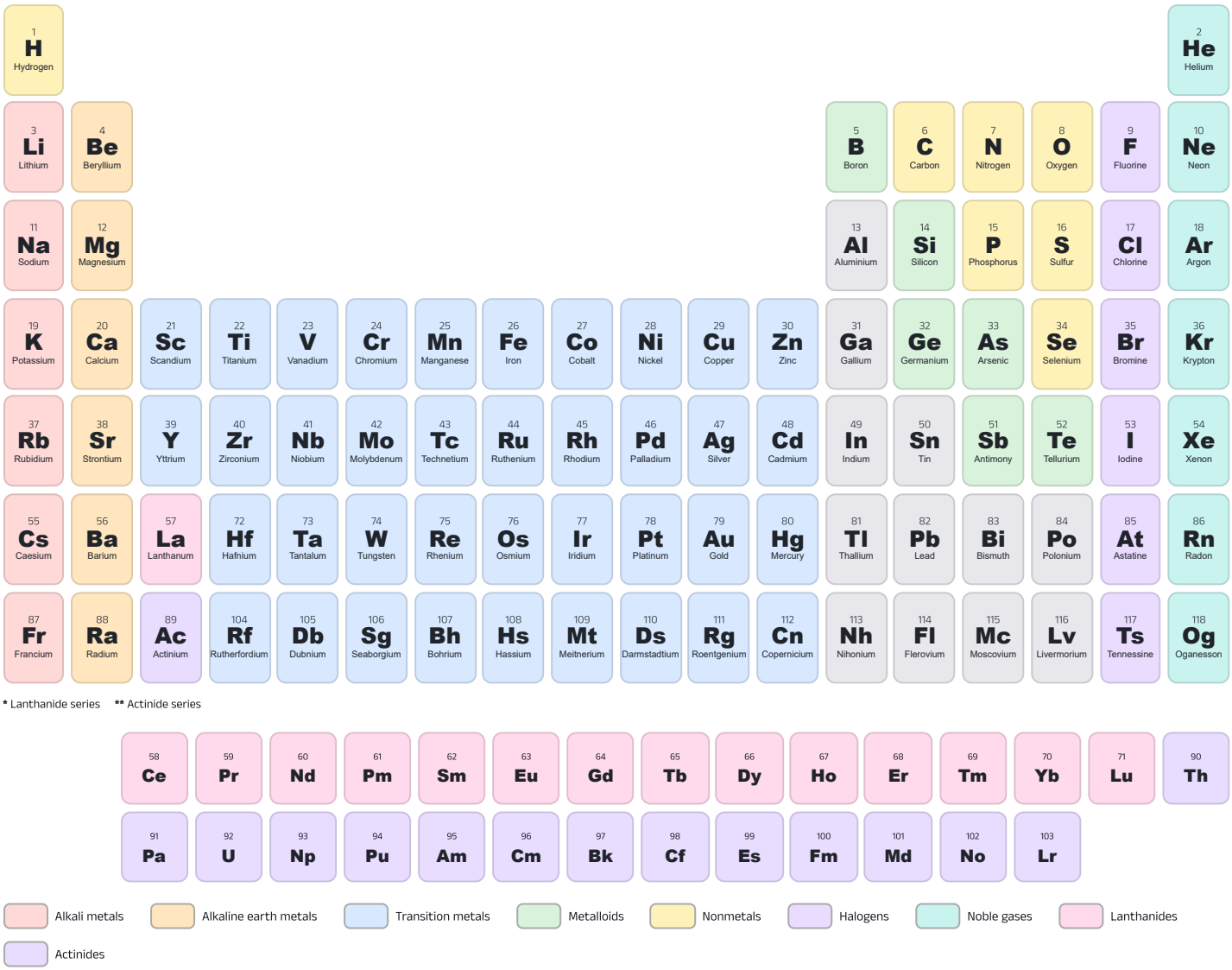
PART 2
BIOLOGY

PART 3
PHYSICS

LEARNING • REVISION • QUESTION & ANSWER

Periodic Table of Elements

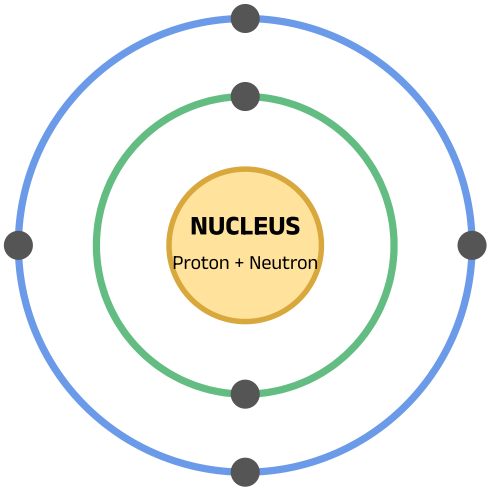
The periodic table arranges the chemical elements according to their atomic numbers and shows important relationships between their properties.



* Lanthanide series ** Actinide series

Remember: Atomic number = Number of protons. In a neutral atom, the number of electrons is equal to the number of protons.

Structure of an Atom



Main Parts of an Atom

An atom has a small central **nucleus** and electrons arranged around it.

- Protons are found in the nucleus.
- Neutrons are found in the nucleus.
- Electrons move around the nucleus in shells.

Subatomic Particles

Particle	Charge	Location
Proton	**Positive (+1e)**	Nucleus
Neutron	**Zero (0)**	Nucleus
Electron	**Negative (−1e)**	Shell

Atomic Number

The atomic number is the number of protons present in the nucleus of an atom.

Atomic number = Number of protons

Mass Number

Mass number is the total number of protons and neutrons in the nucleus.

Mass number = Protons + Neutrons

Electron Shells

Shell	Maximum electrons
K shell	2
L shell	8
M shell	18
N shell	32

Example

Sodium has atomic number 11. Therefore, a sodium atom has 11 protons and, when neutral, 11 electrons.

Its simple shell arrangement is: **2, 8, 1**

PART 3 – CHEMISTRY

Question & Answer – 1

Answer the following questions in simple and clear sentences.

1. What is chemistry?
Answer: Chemistry is the study of matter, its properties and the changes it undergoes.

2. What is matter?
Answer: Matter is anything that has mass and occupies space.

3. Name the three common states of matter.
Answer: Solid, liquid and gas.

4. What is an element?
Answer: An element is a pure substance made of only one type of atom.

5. What is an atom?
Answer: An atom is the smallest unit of an element that retains the properties of that element.

6. What is a molecule?
Answer: A molecule is made when two or more atoms are chemically joined.

7. What is the nucleus of an atom?
Answer: The nucleus is the small central part of an atom containing protons and neutrons.

8. What is the charge of a proton?
Answer: Positive (+1e).

9. What is the charge of an electron?
Answer: Negative (−1e).

10. What is the charge of a neutron?
Answer: Zero (0).

11. Where are protons found?
Answer: Protons are found in the nucleus.

12. Where are neutrons found?
Answer: Neutrons are found in the nucleus.

13. Where are electrons found?
Answer: Electrons are arranged in shells around the nucleus.

14. What is atomic number?
Answer: Atomic number is the number of protons in an atom.

15. What is mass number?
Answer: Mass number is the total number of protons and neutrons.

16. What is a neutral atom?
Answer: A neutral atom has equal numbers of protons and electrons.

17. What is a chemical symbol?
Answer: A chemical symbol is a short form used to represent an element.

18. What is the symbol of hydrogen?
Answer: H.

19. What is the symbol of oxygen?
Answer: O.

20. What is the symbol of carbon?
Answer: C.

21. What is the symbol of sodium?
Answer: Na.

22. What is the symbol of iron?
Answer: Fe.

23. What is the symbol of gold?
Answer: Au.

24. What is the symbol of silver?
Answer: Ag.

25. What is the periodic table?
Answer: The periodic table is an arrangement of chemical elements according to their atomic numbers.

PART 1 – BIOLOGY

The Cell

What is a Cell?

The cell is the basic structural and functional unit of living organisms.

Some organisms consist of one cell, while others consist of many cells.

Important Cell Parts

- **Cell membrane:** controls movement of substances into and out of the cell.
- **Cytoplasm:** jelly-like material inside the cell.
- **Nucleus:** contains genetic material and controls many cell activities.
- **Mitochondria:** release energy from food.
- **Ribosomes:** help make proteins.
- **Vacuole:** stores substances.

Plant and Animal Cells

Plant Cell	Animal Cell
Has cell wall	No cell wall
Usually has chloroplasts	No chloroplasts
Large central vacuole	Small vacuoles may occur

Photosynthesis

Green plants prepare food using sunlight, carbon dioxide and water.

Chlorophyll helps plants absorb light energy.

Respiration

Respiration is the process by which cells release energy from food.

PART 2 – BIOLOGY

Human Body

Digestive System

Breaks food into simpler substances that the body can absorb.

Respiratory System

Takes in oxygen and removes carbon dioxide.

Circulatory System

Transports blood, oxygen, nutrients and wastes.

Nervous System

Controls and coordinates activities of the body.

Skeletal System

Supports the body and protects important organs.

Muscular System

Helps the body move and maintain posture.

Blood

Blood carries oxygen and nutrients and helps transport waste materials. Red blood cells contain haemoglobin, white blood cells help defend the body, and platelets help in blood clotting.

PART 3 – BIOLOGY

Plants, Environment & Question Answer

Plants

- Roots absorb water and minerals.
- Stems support the plant and transport substances.
- Leaves are important for photosynthesis.
- Flowers are involved in reproduction in flowering plants.
- Fruits protect seeds and help in seed dispersal.

Environment

- Air, water and soil are important parts of the environment.
- Plants and animals depend on one another.
- Food chains show the transfer of food energy.
- Pollution can harm living organisms and ecosystems.
- Conservation helps protect natural resources.

1. What is biology?

Answer: Biology is the study of living organisms.

2. What is a cell?

Answer: A cell is the basic structural and functional unit of life.

3. Name two types of cells.

Answer: Plant cells and animal cells.

4. What is the nucleus?

Answer: The nucleus controls many activities of the cell and contains genetic material.

5. What is the cell membrane?

Answer: The cell membrane controls the movement of substances into and out of the cell.

6. What is cytoplasm?

Answer: Cytoplasm is the jelly-like material inside the cell.

7. What is a cell wall?

Answer: A cell wall is a rigid outer layer found in plant cells.

8. What is chlorophyll?

Answer: Chlorophyll is the green pigment that absorbs light energy for photosynthesis.

9. What is photosynthesis?

Answer: Photosynthesis is the process by which green plants prepare food using light energy.

10. Which gas is used in photosynthesis?

Answer: Carbon dioxide.

11. Which gas is released during photosynthesis?

Answer: Oxygen.

12. What is respiration?

Answer: Respiration is the process of releasing energy from food.

13. What is digestion?

Answer: Digestion is the breakdown of food into simpler substances that can be absorbed.

14. Where does digestion begin?

Answer: Digestion begins in the mouth.

15. What is the main function of the stomach?

Answer: The stomach helps digest food.

16. What is the small intestine?

Answer: The small intestine is the main site for digestion and absorption of nutrients.

17. What is blood?

Answer: Blood is a fluid tissue that transports substances around the body.

18. What is the function of red blood cells?

Answer: Red blood cells mainly transport oxygen.

19. What is the function of white blood cells?

Answer: White blood cells help defend the body against harmful organisms.

20. What is the function of platelets?

Answer: Platelets help in blood clotting.

21. What is the heart?

Answer: The heart is a muscular organ that pumps blood around the body.

22. What is the circulatory system?

Answer: It is the body system that transports blood around the body.

23. What is the respiratory system?

Answer: It is the system involved in breathing and gas exchange.

24. What is the main organ of respiration?

Answer: The lungs.

25. What is the nervous system?

Answer: The nervous system controls and coordinates many activities of the body.

PART 3 – BIOLOGY

Question & Answer – 2

1. What is the brain?

Answer: The brain is the main control organ of the nervous system.

2. What is the skeleton?

Answer: The skeleton is the framework of bones that supports the body.

3. What is the function of bones?

Answer: Bones support the body, protect organs and help movement.

4. What are muscles?

Answer: Muscles are tissues that contract and help produce movement.

5. What is an organ?

Answer: An organ is a structure made of different tissues working together.

6. What is a tissue?

Answer: A tissue is a group of similar cells performing a particular function.

7. What is a root?

Answer: A root anchors the plant and absorbs water and minerals from the soil.

8. What is the function of a stem?

Answer: The stem supports the plant and transports water and food.

9. What is the function of a leaf?

Answer: The leaf is an important organ for photosynthesis.

10. What is a flower?

Answer: A flower is the reproductive structure of many flowering plants.

11. What is a seed?

Answer: A seed contains a young plant and stored food for its early growth.

12. What is germination?

Answer: Germination is the process by which a seed begins to grow.

13. What is an ecosystem?

Answer: An ecosystem consists of living organisms interacting with one another and their surroundings.

14. What is a food chain?

Answer: A food chain shows how food and energy pass from one organism to another.

15. What is a producer?

Answer: A producer is an organism, such as a green plant, that makes its own food.

16. What is a consumer?

Answer: A consumer is an organism that obtains food by eating other organisms.

17. What is a decomposer?

Answer: A decomposer breaks down dead organic matter.

18. What is pollution?

Answer: Pollution is the introduction of harmful substances into the environment.

19. Name three types of pollution.

Answer: Air pollution, water pollution and soil pollution.

20. Why are trees important?

Answer: Trees provide oxygen, absorb carbon dioxide, provide habitats and help protect soil.

21. What is conservation?

Answer: Conservation means protecting and wisely using natural resources.

22. What is biodiversity?

Answer: Biodiversity is the variety of living organisms in an area or on Earth.

23. What is adaptation?

Answer: Adaptation is a feature that helps an organism survive in its environment.

24. What is reproduction?

Answer: Reproduction is the process by which organisms produce new organisms.

25. Why is water important for living organisms?

Answer: Water is essential for many body processes and for the survival of living organisms.

PART 1 – PHYSICS

Motion and Force

Motion

Motion is the change in position of an object with respect to time.

Speed tells us how fast an object moves.

Speed = Distance ÷ Time

Force

Force is a push or pull that can change the motion or shape of an object.

The SI unit of force is the **newton (N)**.

Effects of Force

Start Motion

A force can make a stationary object move.

Stop Motion

A force can stop a moving object.

Change Direction

A force can change the direction of motion.

PART 2 – PHYSICS

Pressure, Work and Energy

Pressure

Pressure is the force acting on a unit area.

Pressure = Force ÷ Area

SI unit: **pascal (Pa)**

Work

Work is done when a force causes an object to move through a distance.

Work = Force × Distance

Energy

Energy is the ability to do work.

- Kinetic energy is the energy of motion.
- Potential energy is stored energy.
- Heat, light and electrical energy are common forms of energy.

Simple Formulae

Quantity	Formula	SI Unit
Speed	Distance ÷ Time	m/s
Pressure	Force ÷ Area	Pa
Work	Force × Distance	J

PART 3 – PHYSICS

Light, Sound, Heat & Question Answer

Light

Light enables us to see objects. It travels in straight lines in a uniform medium.

Sound

Sound is produced by vibrating objects and needs a medium to travel.

Heat

Heat is energy transferred from a hotter object to a colder object.

Important Points

- Reflection is the bouncing back of light.
- Refraction is the change in direction of light when it enters another medium.
- Sound cannot travel through a vacuum.
- Temperature tells how hot or cold an object is.
- Heat can be transferred by conduction, convection and radiation.

1. What is physics?
Answer: Physics is the study of matter, energy, motion and the forces acting on objects.

2. What is motion?
Answer: Motion is the change in position of an object with time.

3. What is rest?
Answer: An object is at rest when its position does not change with respect to its surroundings.

4. What is distance?
Answer: Distance is the total length of the path travelled by an object.

5. What is displacement?
Answer: Displacement is the shortest distance from the initial position to the final position in a particular direction.

6. What is speed?
Answer: Speed is the distance travelled per unit time.

7. Write the formula for speed.
Answer: Speed = Distance ÷ Time.

8. What is the SI unit of speed?
Answer: Metre per second (m/s).

9. What is force?
Answer: Force is a push or pull that can change the motion or shape of an object.

10. What is the SI unit of force?
Answer: Newton (N).

11. What can a force do?
Answer: A force can start, stop, speed up, slow down, change direction or change the shape of an object.

12. What is gravity?
Answer: Gravity is the force of attraction between masses.

13. What is friction?
Answer: Friction is a force that opposes motion between surfaces in contact.

14. Give one useful effect of friction.
Answer: Friction helps us walk without slipping.

15. What is pressure?
Answer: Pressure is force acting on a unit area.

16. Write the formula for pressure.
Answer: Pressure = Force ÷ Area.

17. What is the SI unit of pressure?
Answer: Pascal (Pa).

18. Why does a sharp knife cut easily?
Answer: A sharp edge has a small area, so it produces greater pressure for the same force.

19. What is work?
Answer: Work is done when a force causes an object to move through a distance.

20. Write the formula for work.
Answer: Work = Force × Distance.

21. What is the SI unit of work?
Answer: Joule (J).

22. What is energy?
Answer: Energy is the ability to do work.

23. What is kinetic energy?
Answer: Kinetic energy is the energy possessed by a moving object.

24. What is potential energy?
Answer: Potential energy is stored energy due to position or condition.

25. What is heat?
Answer: Heat is energy transferred from a hotter object to a colder object.

PART 3 – PHYSICS

Question & Answer – 2

1. What is temperature?

Answer: Temperature is a measure of how hot or cold an object is.

2. Name three methods of heat transfer.

Answer: Conduction, convection and radiation.

3. What is conduction?

Answer: Conduction is the transfer of heat through a substance without bulk movement of the substance.

4. What is convection?

Answer: Convection is heat transfer by the movement of fluids.

5. What is radiation?

Answer: Radiation is the transfer of heat through electromagnetic waves.

6. What is light?

Answer: Light is a form of energy that enables us to see objects.

7. How does light normally travel through a uniform medium?

Answer: Light normally travels in straight lines.

8. What is reflection?

Answer: Reflection is the bouncing back of light from a surface.

9. What is refraction?

Answer: Refraction is the change in direction of light when it passes from one medium to another.

10. What is a shadow?

Answer: A shadow is a dark region formed when an opaque object blocks light.

11. What is sound?

Answer: Sound is a form of energy produced by vibrating objects.

12. Can sound travel through a vacuum?

Answer: No. Sound needs a material medium to travel.

13. What produces sound?

Answer: Vibrating objects produce sound.

14. What is vibration?

Answer: Vibration is a repeated to-and-fro motion.

15. What is frequency?

Answer: Frequency is the number of vibrations made in one second.

16. What is the unit of frequency?

Answer: Hertz (Hz).

17. What is an electric current?

Answer: Electric current is the flow of electric charge through a conductor.

18. What is the SI unit of electric current?

Answer: Ampere (A).

19. What is a conductor?

Answer: A conductor is a material that allows electric current to pass through it easily.

20. Give one example of a conductor.

Answer: Copper is a good conductor of electricity.

21. What is an insulator?

Answer: An insulator is a material that does not allow electric current to pass through it easily.

22. Give one example of an insulator.

Answer: Rubber is a good electrical insulator.

23. What is a magnet?

Answer: A magnet is an object that attracts certain magnetic materials such as iron.

24. What are the two poles of a magnet?

Answer: North pole and South pole.

25. What happens when like magnetic poles are brought together?

Answer: Like poles repel each other.

Quick Revision: Motion • Force • Pressure • Work • Energy • Light • Sound • Heat